VETRI TECHNOLOGY SOLUTIONS

PYTHON LEARNING MODULES – 60 DAYS

Day 1

Introduction to Web Development + HTML Structure

1. What is Web Development?

Web Development is the process of creating websites and web applications that run on the internet. It involves building everything users see (frontend) and everything that works behind the scenes (backend).

Simple Definition:

Web Development = Designing + Building + Maintaining websites

Types of Web Development

1. Frontend (Client Side)

This is what users see and interact with.

HTML → Structure

CSS → Design

JavaScript → Logic & Interactivity

Example:

Buttons, forms, images, menus, animations

2. Backend (Server Side)

This is what users don't see.

Handles data

Stores information

Processes requests

Example:

Login system, database, payments

3. Full Stack Developer

A person who knows BOTH frontend + backend.

2. What is Client vs Server?

Client

Client is the user's device (browser).

Examples:

Chrome

Firefox

Mobile browser

Client sends request (like opening a website)

Server

Server is a powerful computer that stores website data.
It responds to client requests.

How it works (Simple Flow)

1. User opens browser (Client)
2. User types URL (example: google.com)
3. Client sends request to server
4. Server processes request
5. Server sends response (HTML page)
6. Browser displays website

Example:

When you open YouTube:

Client → Your mobile/laptop

Server → YouTube servers

Response → Video page loads

3. HTML (HyperText Markup Language)

HTML is the foundation of all web pages.

It is used to create structure.

HTMLisNOTaprogramminglanguage.

It is a markup language.

HTML Skeleton (Basic Structure)

```
<!DOCTYPE html>
<html>
<head>
   <title>My First Web Page</title>
</head>

<body>

   <h1>Welcome to Web Development</h1>
   <p>This is my first HTML page.</p>

</body>
</html>
```

Explanation of HTML Structure

1. <!DOCTYPE html>

- Tells browser this is an HTML5 document

2. <html>

- Root element of the page

3. <head>

Contains meta information (not visible)

Examples:

- Title
- SEO data
- CSS links

4. <title>

- Sets browser tab name

5. <body>

- Visible content of webpage

Examples:

Text

Images

Buttons

Real Life Analogy

HTML Part	Real Life Example
<html>	Whole house
<head>	Planning + design blueprint
<body>	Inside rooms (furniture, TV, etc.)

10 Coding Tasks (Day1 Practice)

Task 1 Create a basic HTML page with title "My Website"

Task 2 Display a heading "Welcome to Web Development"

Task 3 Add a paragraph about yourself

Task 4 Create a page with:

- h1
- h2
- p tags

Task 5

Create a webpage showing:

- Your name
- Your course
- Your college/company

Task 6

Create a page with a title "Client Server Model"

Task 7

Write a simple explanation of HTML inside <p> tag

Task 8

Create a structure using:

- html
- head
- body
- title

Task 9 Create a webpage showing "Hello World" in h1 and paragraph below it

Task 10 (Challenge) Create a mini profile page with:

Name (h1)

Bio (p)

Qualification (p)

Goal (p)

Mini Projects

Project 1: Simple Profile Page

Requirements:

- Name
- About yourself
- Skills
- Contact info

Use only HTML tags (h1, p, etc.)

Project 2: Web Development Info Page

Sections:

- What is Web Development
- Frontend explanation
- Backend explanation

Project 3: Student Information Page

Include:

- Student Name
- Class
- Department
- College name

Project 4: Basic Article Page

Create a simple article layout:

- Title (h1)
- Subtitle (h2)

- Content (multiple paragraphs)
- Conclusion

Day 2

Basic HTML Tags (Headings, Paragraphs, Lists, Anchor, Image)

1. HTML Basic Tags Overview

HTML tags are the building blocks of web pages. Each tag has a specific purpose like displaying text, links, images, or lists.

Today we focus on:

- Headings
- Paragraphs
- Lists
- Anchor (links)
- Image

2. Headings in HTML

Headings are used to define titles and sub-titles.

Syntax:

```
<h1>Main Heading</h1>
<h2>Sub Heading</h2>
<h3>Section Heading</h3>
<h4>Smaller Heading</h4>
<h5>Very Small Heading</h5>
<h6>Smallest Heading</h6>
```

Key Points:

- <h1> is the biggest and most important
- <h6> is the smallest
- Used for SEO (Search Engine Optimization)

Example:

```
<h1>Welcome to My Website</h1>
<h2>About Me</h2>
<h3>My Skills</h3>
```

3. Paragraph Tag

Used to write normal text content.

Syntax:

```
<p>This is a paragraph.</p>
```

Example:

<p>I am learning HTML step by step.</p>
<p>This is my second paragraph.</p>

Important:

- Paragraph automatically adds spacing
- Used for descriptions and content blocks

4. Lists in HTML

Lists are used to group items.

1. Ordered List (Numbered List)

```
<ol>
   <li>HTML</li>
   <li>CSS</li>
<li>JavaScript</li>
</ol>
```

Output:

1. HTML
2. CSS
3. JavaScript

2. Unordered List (Bullet List)

```
<ul>
   <li>Apple</li>
   <li>Banana</li>
   <li>Mango</li>
</ul>
```

Output:

Apple

Banana

Mango

3. List Item

Item

Real Use Cases:

- Menus
- Navigation bars
- Product lists
- Skill lists

5. Anchor Tag (Links)

Used to create hyperlinks between pages or websites.

Syntax:

Click Here

Example:

Go to Google

Open in New Tab:

Open Google

Important Attributes:

- href → link address
- target="_blank" → open in new tab

6. Image Tag

Used to display images in a webpage.

Syntax:

<img src="image.jpg" alt="description">

Example:

<img src="nature.jpg" alt="Beautiful Nature">

Attributes:

Src

Path of image

Alt

Text shown if image fails

Width & Height:

<img src="nature.jpg" alt="Nature" width="300" height="200">

Real Use Cases:

- Profile pictures
- Product images
- Banners
- Gallery

Tasks (Day 2 Practice)

Task 1

Create a page with 3 headings (h1, h2, h3)

Task 2

Write a paragraph about your favorite hobby

Task 3

Create an ordered list of 5 programming languages

Task 4

Create an unordered list of 5 fruits

Task 5

Create a webpage with:

- Name (h1)
- About (p)
- Skills list (ul)

Task 6

Create a link to YouTube using anchor tag

Task 7

Create a link that opens in a new tab

Task 8

Display an image with alt text

Task 9

Create a page with:

- Heading
- Paragraph
- List
- Link

Task 10 (Challenge)

Create a “My Profile Page” with:

- Name (h1)
- Bio (p)
- Skills (ul)
- Favorite websites (links)
- Profile image

Mini Projects

Project 1: Personal Profile Page

Include:

- Name
- About
- Skills list
- Social links

Project 2: Study Topics Page

Include:

- Headings for subjects
- Lists of topics
- Paragraph explanations

Project 3: Favorite Foods Page

Include:

- Food names (ul list)

- Description (p)
- Image of each food

Project 4: Simple Navigation Page

Include:

- Home
- About
- Contact links (anchor tags)

Day 3

Forms & Tables in HTML

1. HTML Forms – Introduction

Forms are used to collect user data on websites.

Examples:

- Login page
- Registration form
- Contact form
- Checkout page (eCommerce)

Basic Form Structure

```
<form>
  <!-- form elements go here -->
</form>
```

2. Input Types in HTML

Input tags are used to take user input.

Text Input

```
<input type="text" placeholder="Enter your name">
```

Password Input

```
<input type="password" placeholder="Enter password">
```

Email Input

```
<input type="email" placeholder="Enter email">
```

Number Input

```
<input type="number" placeholder="Enter age">
```

Radio Button (Single Choice)

```
<input type="radio" name="gender"> Male
<input type="radio" name="gender"> Female
```

Checkbox (Multiple Choice)

```
<input type="checkbox"> HTML
<input type="checkbox"> CSS
<input type="checkbox"> JS
```

Submit Button

```
<input type="submit" value="Submit">
```

Button

```
<button>Click Me</button>
```

Textarea (Multi-line input)

```
<textarea placeholder="Enter your message"></textarea>
```

Example Form

```
<form>
   <input type="text" placeholder="Name"><br><br>
   <input type="email" placeholder="Email"><br><br>
   <input type="password" placeholder="Password"><br><br>
   <input type="submit" value="Register">
</form>
```

3. HTML Form Validation Basics

Validation means checking user input before submitting.

Required Field

```
<input type="text" required>
```

User must fill this field

Email Validation

```
<input type="email" required>
```

Must enter valid email format

Min & Max (Numbers)

```
<input type="number" min="1" max="100">
```

Minlength & Maxlength

```
<input type="text" minlength="3" maxlength="10">
```

Example Form with Validation

```
<form>
   <input type="text" placeholder="Name" required><br><br>
   <input type="email" placeholder="Email" required><br><br>
   <input type="number" min="18" max="60" placeholder="Age"><br><br>
   <input type="submit" value="Submit">
</form>
```

4. HTML Tables

Tables are used to display data in rows and columns.

Basic Table Structure

```
<table border="1">
   <tr>
      <th>Name</th>
      <th>Age</th>
      <th>City</th>
   </tr>

   <tr>
      <td>John</td>
      <td>25</td>
      <td>Chennai</td>
   </tr>
</table>
```

Explanation

<table>

Creates table

<tr>

Table Row

<th>

Table Header (bold)

<td>

Table Data

Multiple Rows Example

```
<table border="1">
   <tr>
      <th>Name</th>
      <th>Course</th>
      <th>Marks</th>
   </tr>
   <tr>
      <td>Alice</td>
      <td>HTML</td>
      <td>85</td>
   </tr>
```

```
  <tr>
    <td>Bob</td>
    <td>CSS</td>
    <td>90</td>
  </tr>
</table>
```

Tasks (Day 3 Practice)

Task 1 Create a form with:

- Name input
- Email input
- Submit button

Task 2

Create a password input field

Task 3

Create a radio button group for gender

Task 4

Create checkbox list for hobbies

Task 5

Create a contact form with textarea

Task 6

Create a form with validation (required fields)

Task 7

Create a number input with min and max age (18–60)

Task 8

Create a simple table with 3 columns and 3 rows

Task 9

Create a student marks table (Name, Subject, Marks)

Task 10 (Challenge)

Create a “Student Registration Form” with:

- Name
- Email
- Password
- Age
- Gender (radio)
- Skills (checkbox)
- Submit button

Mini Projects

Project 1: Student Registration Form

- Full form with validation
- Gender + skills + submit

Project 2: Contact Us Page

- Name
- Email
- Message (textarea)
- Submit button

Project 3: Student Marks Table System

- Table with multiple students
- Name, Subject, Marks

Project 4: Simple Order Form

- Product name
- Quantity
- Address
- Submit button

Day 4

HTML5 Semantic Tags

1. What are Semantic Tags?

Semantic tags are HTML5 elements that clearly describe their meaning to both:

Browser

Developer

Search engines (SEO)

Insimplewords:

Semantic tags = "Meaningful HTML structure"

Instead of using only <div>, we use meaningful tags like:

- <header>
- <nav>
- <section>
- <article>
- <footer>

Why Semantic Tags are Important?

Better SEO

Search engines understand your content structure easily.

Better Readability

Code becomes clean and understandable.

Better Accessibility

Screen readers can easily interpret content.

2. <header> Tag

Used for top section of a page or section.

Example:

```
<header>
   <h1>My Website</h1>
   <p>Welcome to my homepage</p>
</header>
```

Use cases:

- Website logo
- Title
- Intro text

3. <nav> Tag

Used for navigation links (menu).

Example:

```
<nav>
   <a href="#">Home</a>
   <a href="#">About</a>
   <a href="#">Services</a>
   <a href="#">Contact</a>
</nav>
```

Use cases:

- Menus
- Navigation bars
- Header links

4. <section> Tag

Used to group related content.

Example:

```
<section>
   <h2>About Us</h2>
   <p>We are a web development training institute.</p>
</section>
```

Use cases:

- About section
- Services section
- Content blocks

5. <article> Tag

Used for independent content that can stand alone.

Example:

```
<article>
   <h2>Latest Tech News</h2>
   <p>HTML5 semantic tags improve web structure.</p>
</article>
```

Use cases:

- Blog posts
- News articles
- Product details

6. <footer> Tag

Used for bottom section of webpage.

Example:

```
<footer>
   <p>© 2026 My Website. All rights reserved.</p>
</footer>
```

Use cases:

- Copyright
- Contact info
- Social links

Full Page Example (Semantic Layout)

```
<!DOCTYPE html>
<html>
<head>
   <title>Semantic Web Page</title>
</head>
```

```
<body>

<header>
   <h1>My Website</h1>
</header>
<nav>
   <a href="#">Home</a>
   <a href="#">About</a>
   <a href="#">Contact</a>
</nav>
<section>
   <h2>About Section</h2>
   <p>This is the main content area of the website.</p>
</section>
<article>
   <h2>Latest Blog</h2>
   <p>HTML5 semantic tags make websites more structured.</p>
</article>
<footer>
   <p>© 2026 My Website</p>
</footer>
</body>
</html>
```

Tasks (Day 4 Practice)

Task 1

Create a page with <header> and a title inside it

Task 2

Create a navigation menu using <nav>

Task 3

Create a section with heading and paragraph

Task 4

Create an article about your favorite hobby

Task 5

Create a footer with copyright text

Task 6

Create a webpage using all semantic tags together

Task 7

Create a blog layout using <article> tags

Task 8

Create a website layout with header + nav + section

Task 9

Create a portfolio layout using semantic tags

Task 10 (Challenge)

- Build a full webpage:
- Header (name + tagline)
- Nav menu (4 links)
- Section (about you)
- Article (your achievement)
- Footer (contact info)

4 Mini Projects

Project 1: Personal Website Layout

- Header with name
- Navigation menu
- About section
- Footer

Project 2: Blog Page

- Multiple articles
- Each article with title + content

Project 3: College Website Layout

- Header (college name)
- Navigation (departments, courses)
- Sections (about, admissions)
- Footer

Project 4: Portfolio Structure Page

- Header (your name)
- Sections (skills, projects)
- Article (achievement)
- Footer (contact)